

HW2 protocol

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Task 1-2:

I have started 3 hazelcast nodes using Docker: docker-compose.yml, hazelcast.xml (config file).

```
PS C:\Users\user\Documents\3 course\sem2\apz\hw2> docker compose up -d
time="2026-03-01T18:27:04+02:00" level=warning msg="C:\\Users\\user\\Documents\\3 course\\sem2\\apz\\hw2\\docker-compose.yml: the attribute `version` is obsolete, it
will be ignored, please remove it to avoid potential confusion"
[+] Running 5/5
 ✓ Network hw2 hazelcast-net Created 0.1s
 ✓ Container hz-mc Started 1.6s
 ✓ Container hz2 Started 1.6s
 ✓ Container hz3 Started 1.6s
 ✓ Container hz1 Started 1.6s
PS C:\Users\user\Documents\3 course\sem2\apz\hw2> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
237e88084109   hazelcast/hazelcast:5.3             "hz start"              About a minute ago Up About a minute 0.0.0.0:5702->5701/tcp             hz2
d6c716f75056   hazelcast/hazelcast:5.3             "hz start"              About a minute ago Up About a minute 0.0.0.0:5701->5701/tcp             hz1
db6ce06c05f6   hazelcast/management-center:5.3     "bash ./bin/mc-start..." About a minute ago Up About a minute 8081/tcp, 0.0.0.0:8080->8080/tcp, 8443/tcp hz-mc
4f57b6d2c03d   hazelcast/hazelcast:5.3             "hz start"              About a minute ago Up About a minute 0.0.0.0:5703->5701/tcp             hz3
PS C:\Users\user\Documents\3 course\sem2\apz\hw2>
```

☐	☐	welcome-to-docker	1743d0e3b47	docker/welcome-to-dock	8088:80	0%	1 year ago			
☐	☐	hw1	-	-	-	0%	8 days ago			
☐	☒	hw2	-	-	-	3.58%	26 minutes ago			
☐	☒	hz3	4f57b6d2c03d	hazelcast/hazelcast:5.3	5703:5701 ↗	0.84%	26 minutes ago			
☐	☒	hz2	237e88084109	hazelcast/hazelcast:5.3	5702:5701 ↗	0.81%	26 minutes ago			
☐	☒	hz-mc	db6ce06c05f6	hazelcast/management-c	8080:8080 ↗	1.2%	26 minutes ago			
☐	☒	hz1	d6c716f75056	hazelcast/hazelcast:5.3	5701:5701 ↗	0.73%	26 minutes ago			

HAZELCAST

dev

5.3

Cluster Connections

Docs

SQL Browser

Console

Settings

Dev Mode

5.3.4 - 2026/03/01 - 06:53 PM

Cluster

Dashboard

Clients

Client Filtering - OFF

Members

Administration

WAN Replication

Scripting

Healthcheck

Storage

Stream Processing

Compute

Monitoring

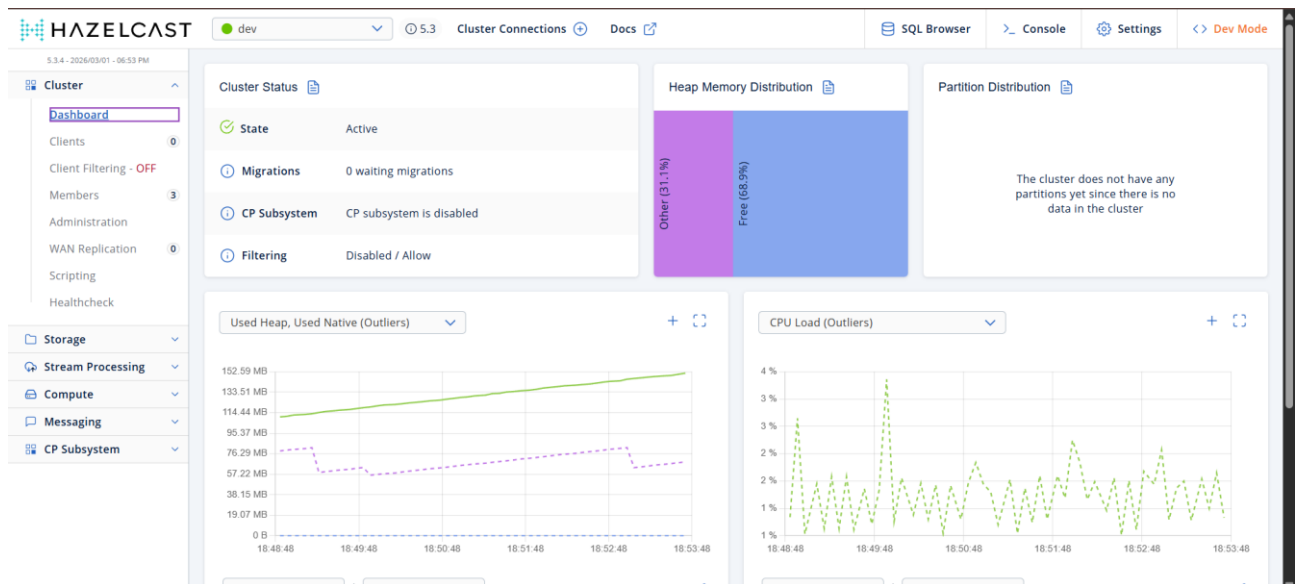
Members

Search

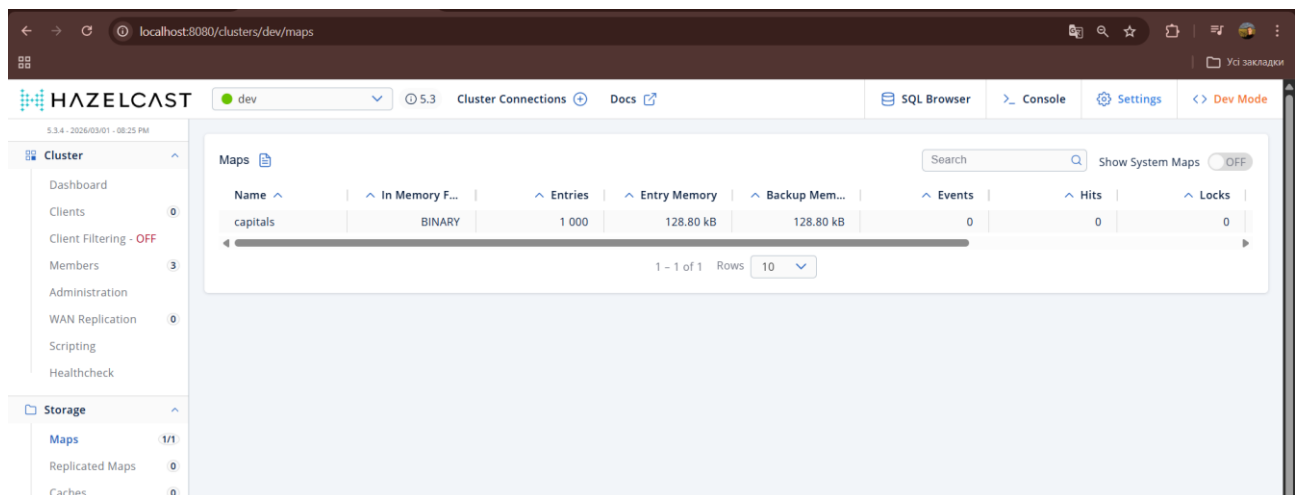
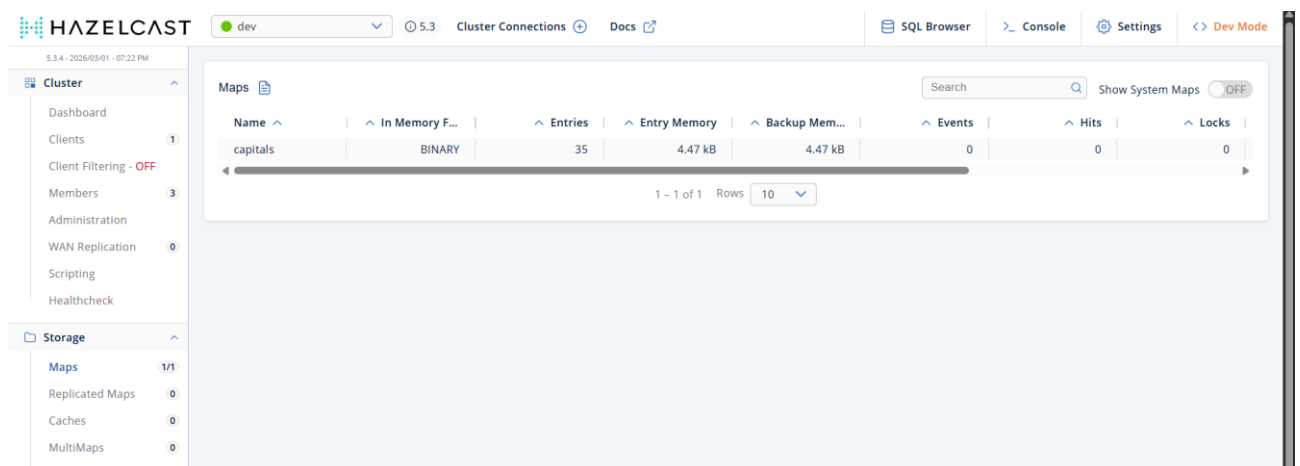
Default View

Member	Addi...	Scrip...	Cons...	Strea...	Slow ...	Haze...	Owned Partitions	OS Committed V...	OS C...
hz1:5701	No Info	Disabled	Disabled	Enabled	No	5.3.8	0	7.18 GB	1.54 %
hz2:5701	No Info	Disabled	Disabled	Enabled	No	5.3.8	0	7.18 GB	1.50 %
hz3:5701	No Info	Disabled	Disabled	Enabled	No	5.3.8	0	7.18 GB	1.54 %

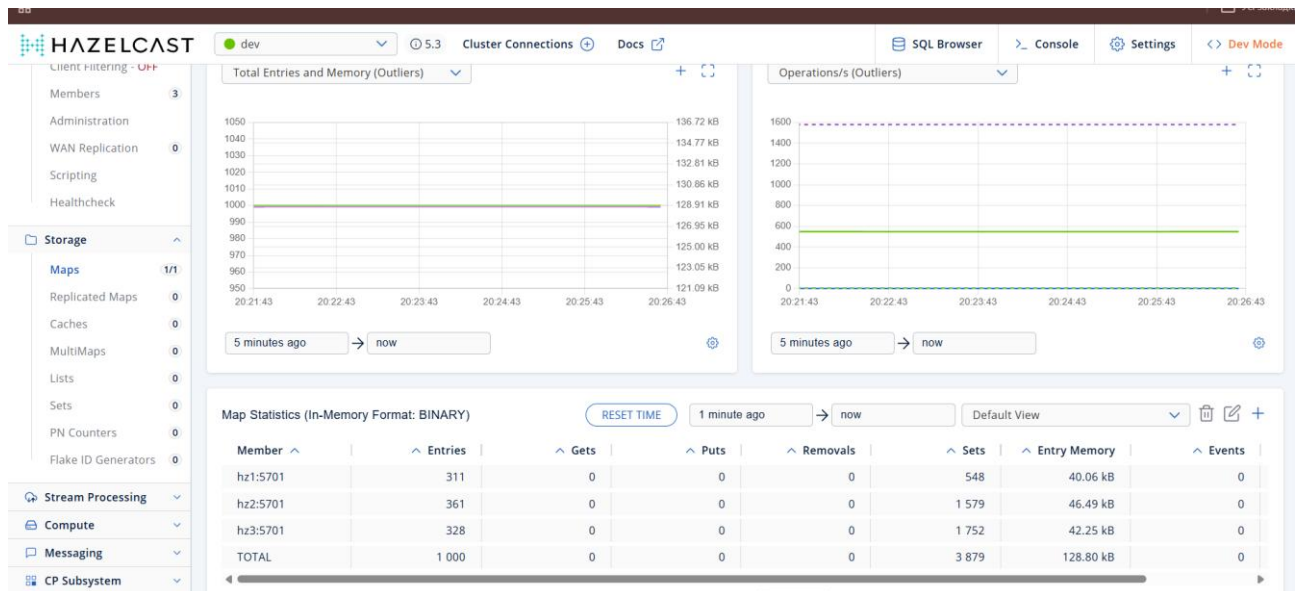
1 - 3 of 3 Rows 10



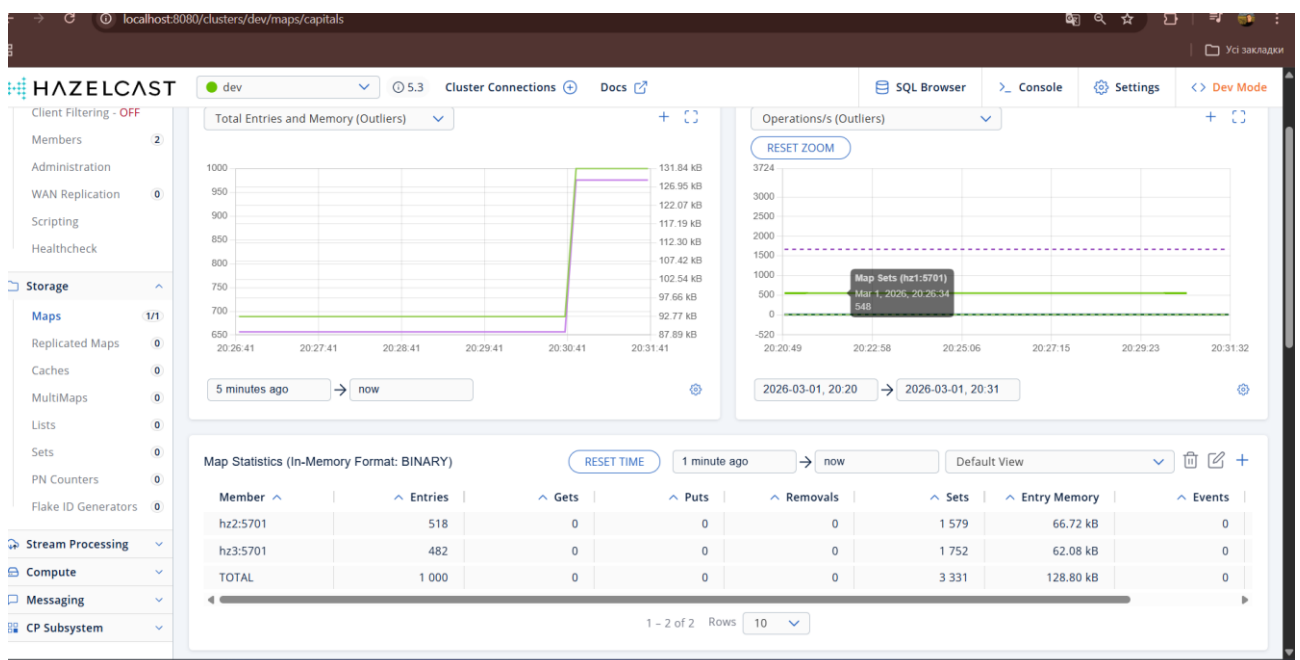
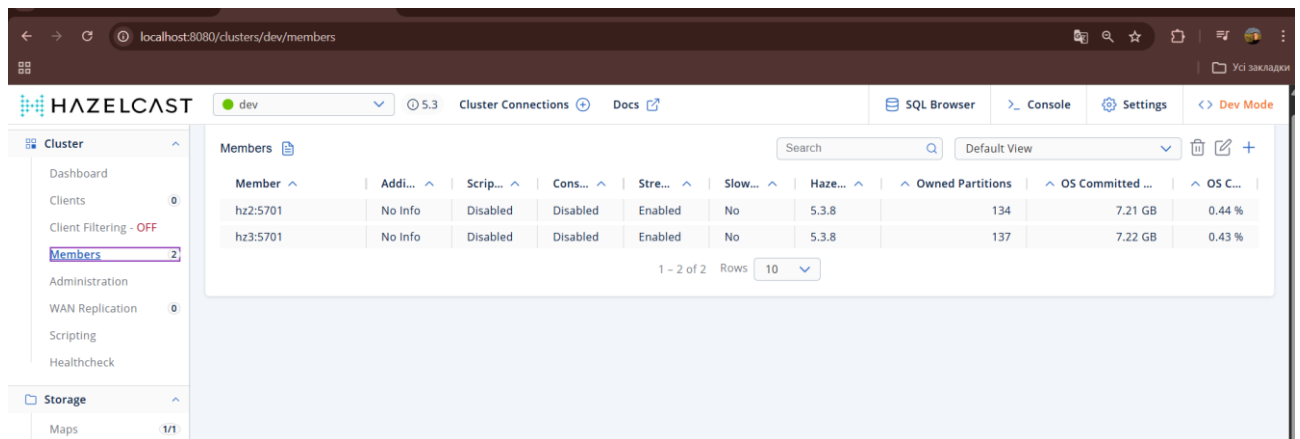
Task 3:



Nodes distribution:



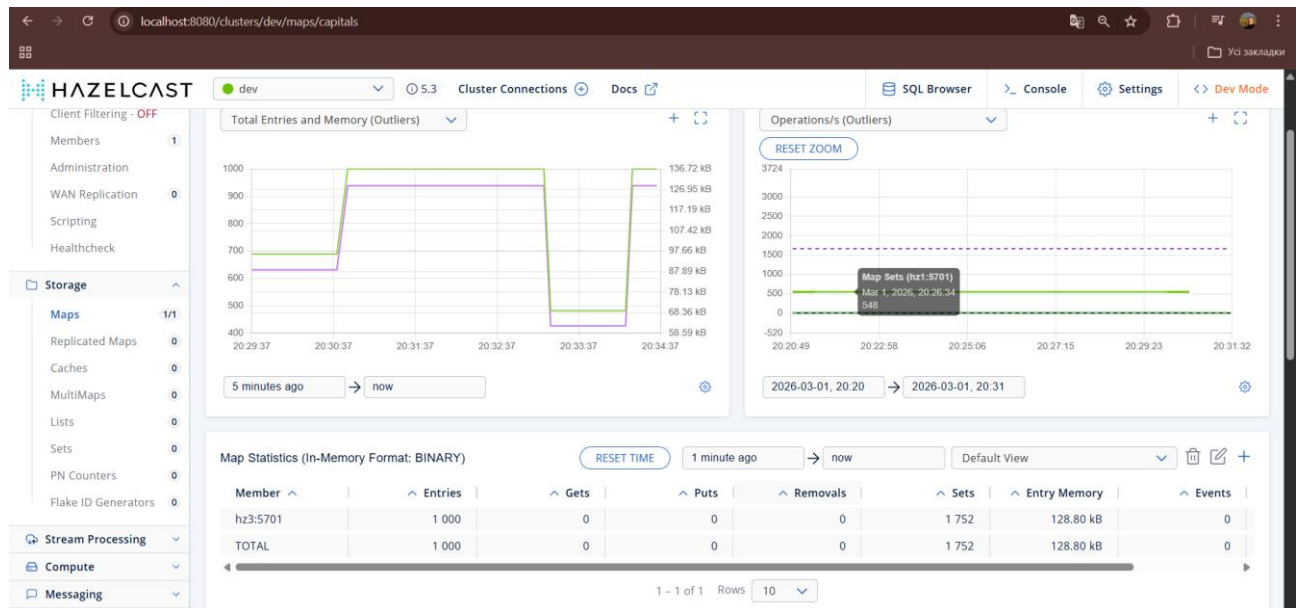
I have run `docker stop hz1:`



Hazelcast distribute data between hz2 (518) and hz3 (482)

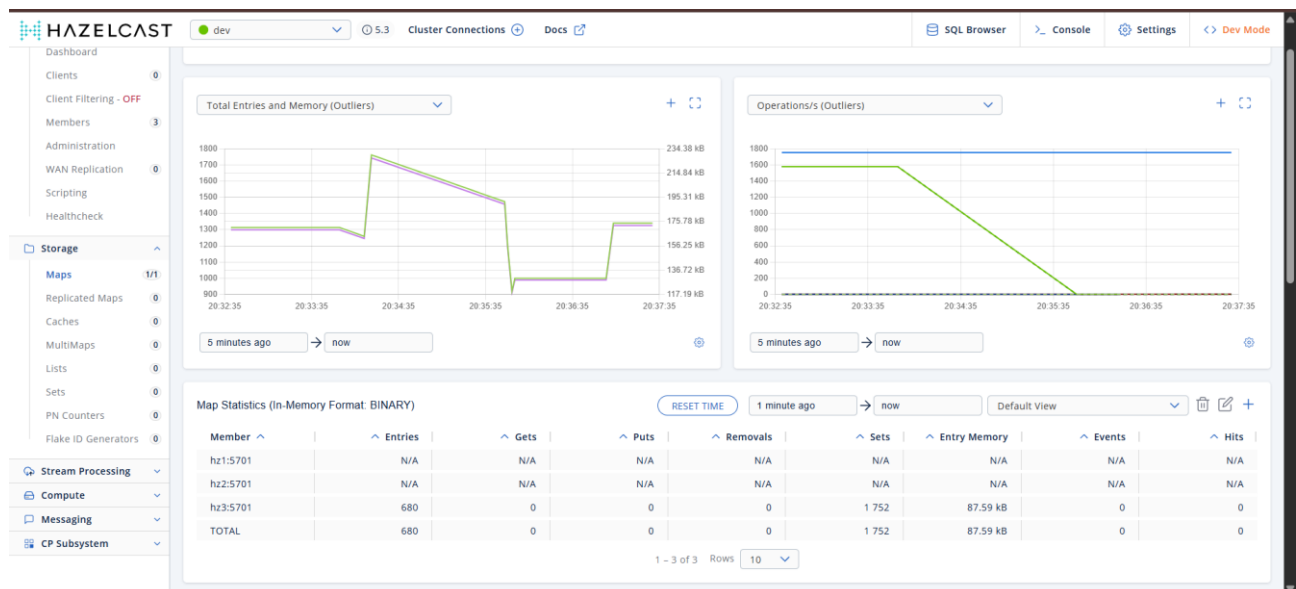
Next, I have disabled the second node (hz2):

Now all the data is stored on a single node (hz3):



Next step: I have recovered disabled nodes.

When you killed hz1 and hz2 simultaneously, the cluster suffered immediate data loss, with the total entry count dropping from 1,000 to 680. Because both the primary data and its only backup were stored across the two failed nodes, the remaining member (hz3) was unable to recover the missing partitions.



To prevent this problem, you need to increase the number of backups. For a 3-node cluster, setting the `<backup-count>2</backup-count>` parameter in the hazelcast.xml file ensures that each partition has copies on all nodes. In this case, the data will survive even if 2 nodes fail simultaneously.

Task 4

3 processes are processing simultaneously

The screenshot shows the Hazelcast Dev UI interface. The left sidebar contains navigation links for Cluster, Storage, Stream Processing, Compute, and Messaging. The main content area is titled 'Clients' and shows a table of 4 clients. The table has columns for Name, Address, Type, Member Connection, Hazelcast Client Version, and UUID. The clients are:

Name	Address	Type	Member Connection	Hazelcast Client Version	UUID
hz.client_0	172.20.0.1	Python	N/A	5.6.0	44520e39-acf3-49a1-a1e8-4cf...
MC-Client-dev	hz-mc.hw2_hazelcast-net	Management Center	ALL	5.3.8	b4c4b35a-c6bc-49d7-b3d7-d3...
hz.client_0	172.20.0.1	Python	N/A	5.6.0	4db46f8c-350c-47df-b177-6d2...
hz.client_0	172.20.0.1	Python	N/A	5.6.0	3fd37527-5b08-4b9c-9485-7e...

The table shows 1 - 4 of 4 Rows. The pagination dropdown is set to 10.